



1 Problem Statement & Market Friction

Students, developers, and knowledge workers regularly rely on general-purpose AI chatbots (ChatGPT, Gemini, Claude) for academic and engineering workflows. However, a single generic chatbot fails when tasked with fundamentally disparate domains:

- **Learning Inefficiency:** Generic chatbots dump complete answers rather than guiding students step-by-step using interactive Socratic pedagogy.
- **Unstructured Code Feedback:** Standard chat produces monolithic text. Real engineering reviews require categorized, actionable feedback (Bugs, Security, Performance, Style).
- **Document Hallucinations:** Standard LLMs hallucinate external training knowledge into document Q&A instead of adhering strictly to verified text.
- **Vague Career Advice:** Generic resume critiques lack actionable, quantified before/after rewrites aligned with modern ATS standards.
- **Single-Provider Fragility:** Reliance on a single LLM API leaves users stranded during quota exhaustion (HTTP 429), regional outages, or degradation.

Klyvix resolves this by delivering a unified, production-hardened platform housing **6 specialized AI personas**, intelligent multi-LLM failover routing, and dense vector retrieval.

2 Proposed Solution & Architecture Stack

Klyvix is engineered as a single-container full-stack platform containerized on **Google Cloud Run** with zero-overhead async streaming and modular persona orchestration.

Architecture Layer	Selected Technology	Engineering Rationale & Production Role
Backend Engine	FastAPI (Python 3.12) + Uvicorn	Native async concurrency, Server-Sent Events (SSE) streaming, automated OpenAPI generation.
Frontend Interface	Vanilla HTML5 / CSS3 / ES6+ JS	Zero npm build step, sub-50ms render, custom dark "Elite Terminal" styling with Marked.js & KaTeX.
Primary Reasoning LLM	Google Gemini 2.0 Flash	High-quality multi-modal reasoning, native Google Search grounding, generous free-tier throughput.
Specialized Code LLM	Mistral Codestral	Deep code intelligence, surgical refactoring, and strict compliance with structured JSON schemas.
Fallback Inference LLM	Groq Llama 3.3 70B	Sub-300ms LPU's providing seamless automated failover when Gemini/Mistral encounters rate limits.
Vector Database (RAG)	Qdrant Cloud (768-dim)	Cosine similarity vector search over dense Gemini embeddings for strict document-grounded Q&A.
Relational Database	PostgreSQL (Neon Serverless)	Persistent cloud storage for user accounts, uploaded document metadata, and chat sessions.
Caching & Rate Limiting	Redis (Upstash Cloud)	Distributed token-bucket rate limiting and LLM response caching to protect upstream quotas.
Cloud Deployment	Google Cloud Run (Docker)	Serverless auto-scaling (0-10 instances), auto-HTTPS, TLS 1.3, hosted at europe-west1.

Core Architecture Principle: Personas as Data (Configuration Objects)

Personas are data, not separate microservices. A single high-efficiency streaming pipeline (/api/chat/{persona_id}) serves all six personas — dynamically binding system prompts, output schemas, temperature parameters, and retrieval pipelines.

3 The Six Specialized AI Personas

Each persona in Klyvix is engineered with specialized behavioral constraints, schema enforcement, and tailored prompting:

 **1. Study Mentor**

Socratic • Direct

Personalized academic tutor. Offers **Direct Mode** for structured, exam-ready explanations with LaTeX math and comparison tables, and **Socratic Mode** for step-by-step guided questioning that verifies mastery before progressing.

 **2. Code Reviewer**

Structured JSON

Performs rigorous automated code audits across four distinct dimensions: **Bugs & Correctness**, **Security**, **Performance**, and **Style**. Emits validated JSON parsed into interactive, color-coded UI cards.

 **3. Code Colleague**

Pair Programming

Senior engineer pair-programming assistant. Generates production-ready code with inline comments, performs surgical non-destructive refactoring, explains architecture trade-offs, and clarifies ambiguous edge cases.

 **4. Document Analyzer**

Qdrant RAG • 0% Hallucination

Processes uploaded PDFs via semantic chunking and 768-dimensional Qdrant vector retrieval. Answers queries *strictly* from retrieved document chunks with exact citations, explicitly refusing to extrapolate or hallucinate.

 **5. Resume Reviewer**

ATS Audit • JSON

Evaluates technical resumes against industry hiring benchmarks. Categorizes findings into **Strengths**, **Gaps**, and **Actionable Suggestions** with concrete before-and-after bullet rewrites and quantified metrics.

 **6. Research Assistant**

Multi-Source • Search Grounded

Synthesizes academic papers and real-time Google Search data into structured thematic literature reviews. Identifies consensus vs. controversy, flags source conflicts, and outputs bibliographic citations.

4 Key Technical Features & Engineering Innovations

 **Real-Time SSE Token Streaming**

Server-Sent Events (SSE) stream token-by-token directly to the DOM. Time-to-First-Token (TTFT) averages under 1.5 seconds, providing an instantaneous terminal chat experience without websocket connection overhead.

 **Multi-LLM Failover Engine**

Backend error traps intercept Gemini/Mistral rate limits (HTTP 429) or network timeouts and seamlessly hot-swap to Groq Llama 3.3 70B without severing client streaming or showing error toasts.

 **Dense Semantic Vector Pipeline**

Automated PDF text extraction via pypdf, dynamic 400-word overlap chunking, Gemini text-embedding-004 vector generation, and Qdrant Cloud indexing with cosine threshold filtering.

 **Production Auth & Security**

Stateless JWT session tokens, bcrypt password hashing, strict CORS origin locking, and environment-isolated API credentials ensuring enterprise-grade isolation.

5 Competitive Innovation & Differentiation Matrix

Feature Dimension	Generic Consumer Chatbot	Klyvix AI Command Center
Persona Architecture	1 generic persona with superficial roleplay prompts	6 dedicated personas with custom schemas, tools & parameters
Output Modalities	Unstructured plain text / standard markdown	Structured JSON UI cards , KaTeX math formulas, syntax blocks
Document Grounding	High hallucination risk; mixes external training weights	Strict Qdrant vector retrieval with chunk-level citations & zero hallucination
Inference Reliability	Single provider (breaks on rate-limits/outages)	Multi-LLM Router with automated Gemini → Mistral → Groq failover
User Experience	Generic consumer chat bubbles	Cyberpunk/Elite Terminal UI with keyboard shortcuts & zero build delay
Pedagogical Method	Dumps answers immediately	Dual Mode: Direct structured synthesis or Socratic step-by-step guidance

6 Live Deployment Verification & Deliverables

Klyvix is fully built, tested, and operational in production on Google Cloud Run:



Production Application

Live URL: <https://klyvix-177846015206.europe-west1.run.app/>
Hosted on Google Cloud Run with automatic SSL certificate management and worldwide low-latency routing.



API Documentation

Swagger UI: <https://klyvix-177846015206.europe-west1.run.app/docs>
Interactive OpenAPI interface for testing streaming endpoints, file uploads, and session management.

7 Expected Outcomes & Milestone Deliverables

- **Six Purpose-Built AI Personas:** Demonstrating demonstrable prompt engineering differentiation across learning, code review, pair programming, document RAG, resume audit, and multi-source research.
- **Intelligent Multi-LLM Failover:** Production-verified uptime and zero-interruption user experience during upstream rate limits.
- **Elite Terminal Interface:** Fast, responsive, dark-mode aesthetic with live KaTeX math and Highlight.js code rendering.
- **End-to-End Cloud Deployment:** Containerized Docker build with automatic CI/CD deployment on Google Cloud Run.